

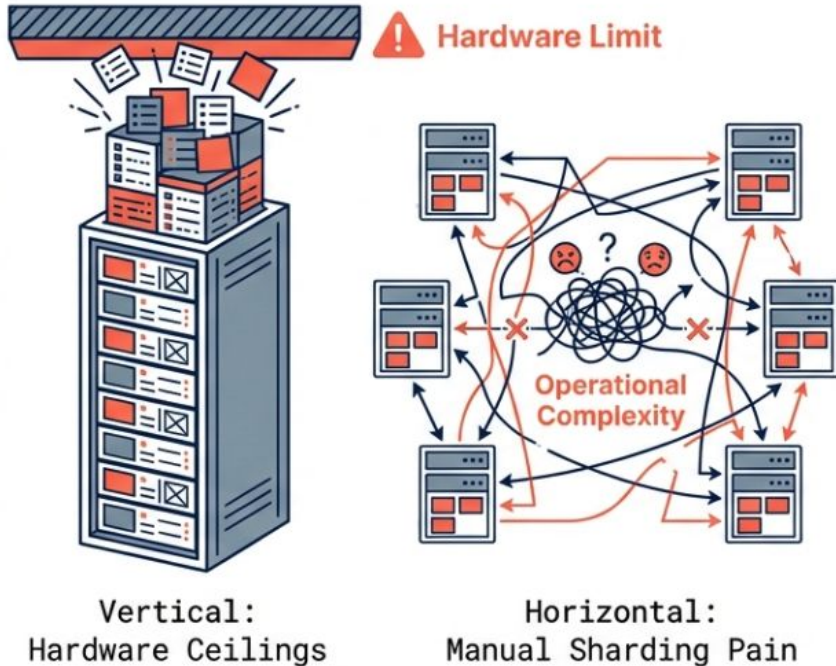


# TiDB

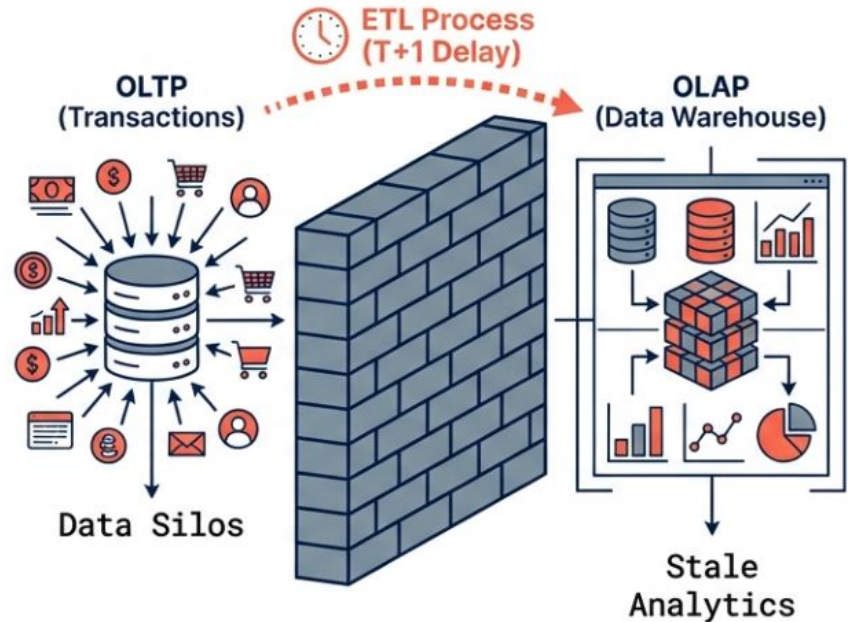
## Modern Database Architecture for Real-Time Workloads

# Why new database

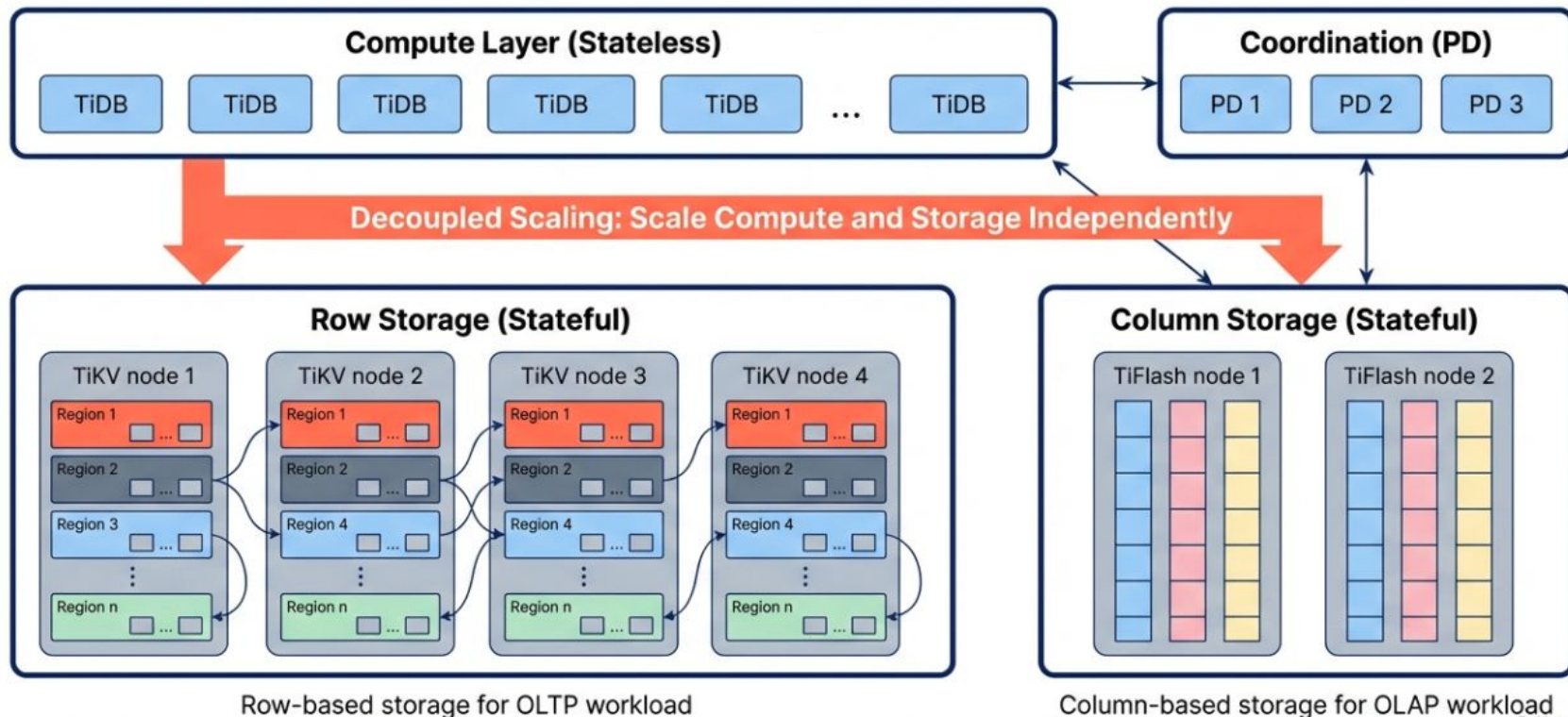
## The Scaling Problem



## The Workload Wall

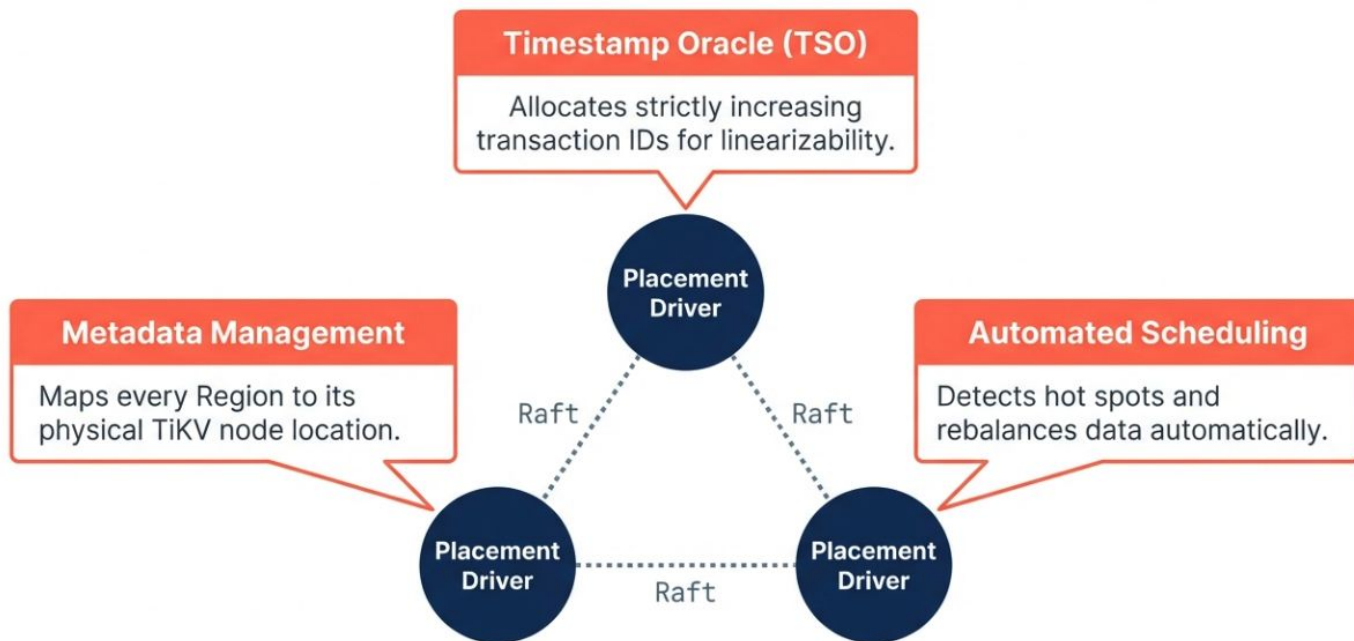


# Architecture



# Placement Driver (PD)

## The Coordinator (PD): The Cluster's Nervous System



# TiDB Server

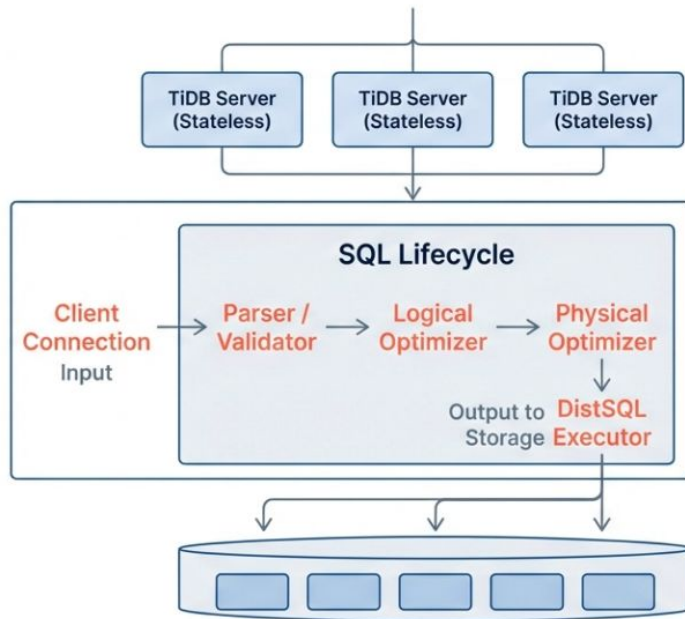
## The SQL Layer (TiDB Server): The Stateless Brain

**Protocol:** MySQL 5.7 Compatible

**Parsing & Compiling:** AST  
Generation

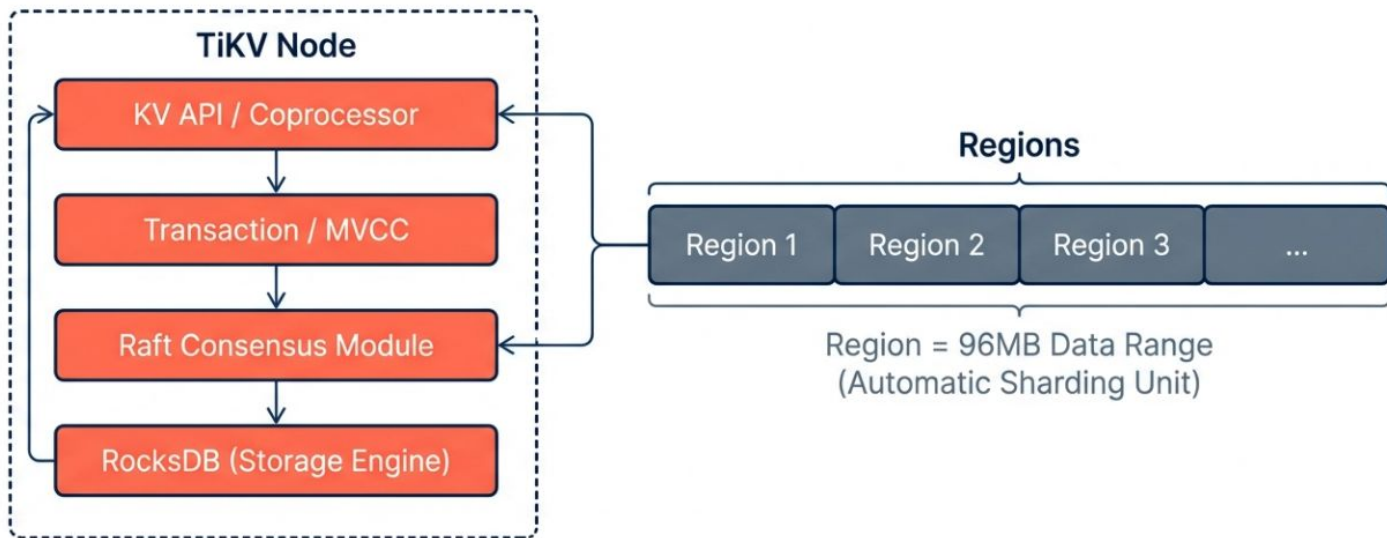
**Translation:** Relational SQL →  
Key-Value Requests

**Scaling:** Stateless (Add nodes  
instantly)



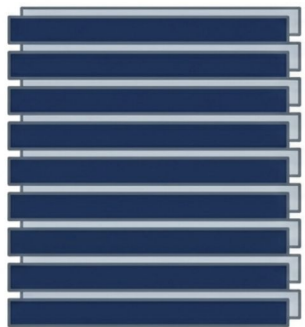
# TiKV

## The Storage Layer (TiKV): The Distributed Heart

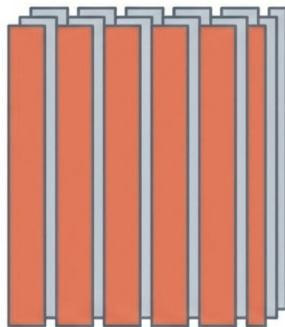


# TiFlash

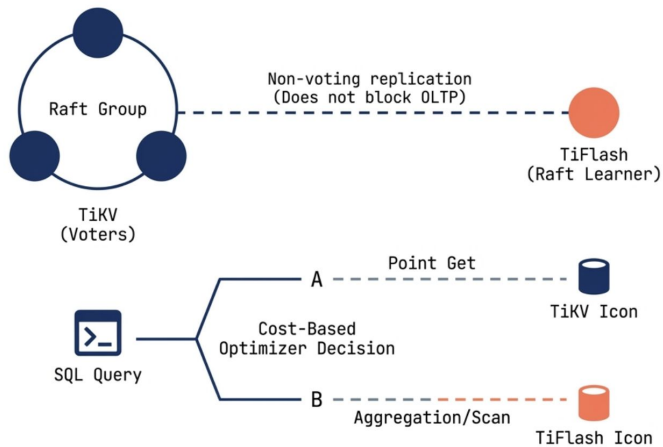
TiKV (Row Store)



TiFlash (Column Store)



Real-time Asynchronous  
Replication.



## Who is using it

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**CAPCOM®**

 **PLAID**

 **Square**

**Bolt**

**Trip.com**

 **Pinterest**



# Trade-offs



## ✓ Pros

- Horizontal Scaling
- HTPA
- Good documentation
- Build modern language, Go, Rust
- K8S operator
- TiCDC
- TiDB Cloud

## ✗ Cons

- Complex infra setup
- Hostspots
- Advanced configuration for throughput (SHARD\_ROW\_ID\_BITS)
- Write latency
- No triggers and stored procedures



T H E  
E N D

# Comparisons



| Feature                       | TiDB                               | MySQL + Vitess               | YugabyteDB                        | CockroachDB                     |
|-------------------------------|------------------------------------|------------------------------|-----------------------------------|---------------------------------|
| SQL Compatibility             | MySQL                              | MySQL                        | PostgreSQL                        | PostgreSQL                      |
| Horizontal Scaling            | Native automatic scaling           | Sharding-based scaling       | Native automatic scaling          | Native automatic scaling        |
| Distributed ACID Transactions | Yes                                | Limited / complex            | Yes                               | Yes                             |
| Sharding                      | Automatic                          | Manual / configured          | Automatic                         | Automatic                       |
| Multi-Region Support          | Yes                                | Difficult to implement       | Yes                               | Very strong                     |
| Consistency Model             | Strong consistency                 | Depends on configuration     | Strong consistency                | Strong consistency              |
| Architecture                  | SQL layer + distributed KV storage | Proxy/router + MySQL shards  | PostgreSQL layer + distributed KV | Integrated SQL + distributed KV |
| Storage Engine                | TiKV                               | InnoDB (MySQL)               | DocDB                             | RocksDB                         |
| OLTP Workloads                | Excellent                          | Excellent                    | Excellent                         | Excellent                       |
| OLAP / Analytics              | Good (TiFlash column store)        | Weak                         | Moderate                          | Moderate                        |
| Operational Complexity        | Medium                             | High                         | Medium                            | Medium                          |
| Ecosystem                     | Growing                            | Very large (MySQL ecosystem) | Medium                            | Large                           |
|                               |                                    |                              |                                   |                                 |